Semantic Clustering and Lexical Retrieval in Hindi Bilinguals: Evidence from Verbal Fluency and Spatial Arrangement Tasks

BRSM PROJECT POSTER PRESENTATION SP 2026

INTRODUCTION

How do you find the right word in your mind? When asked to name as many animals as you can in 60 seconds, your mind walks through related ideas (cat → dog → wolf → tiger). We study this "mental walk" in 35 Hindi-English bilingual speakers via two tasks:

- **Verbal Fluency Task (VFT):** Name as many [category] as you can in 60 seconds – every word and inter-word time recorded.
- **Spatial Arrangement Method (SpAM):** Drag the same words on a blank canvas; closer = more related. The 2-D layout reveals each person's mental map. The link between SpAM distances and VFT timing reveals the geometry of mental search. Hindi – with two scripts (Devanagari + Roman) and rich code-mixing – lets us test which findings are universal vs. language-specific. Verbal fluency is also used clinically to screen for Alzheimer's and depression.

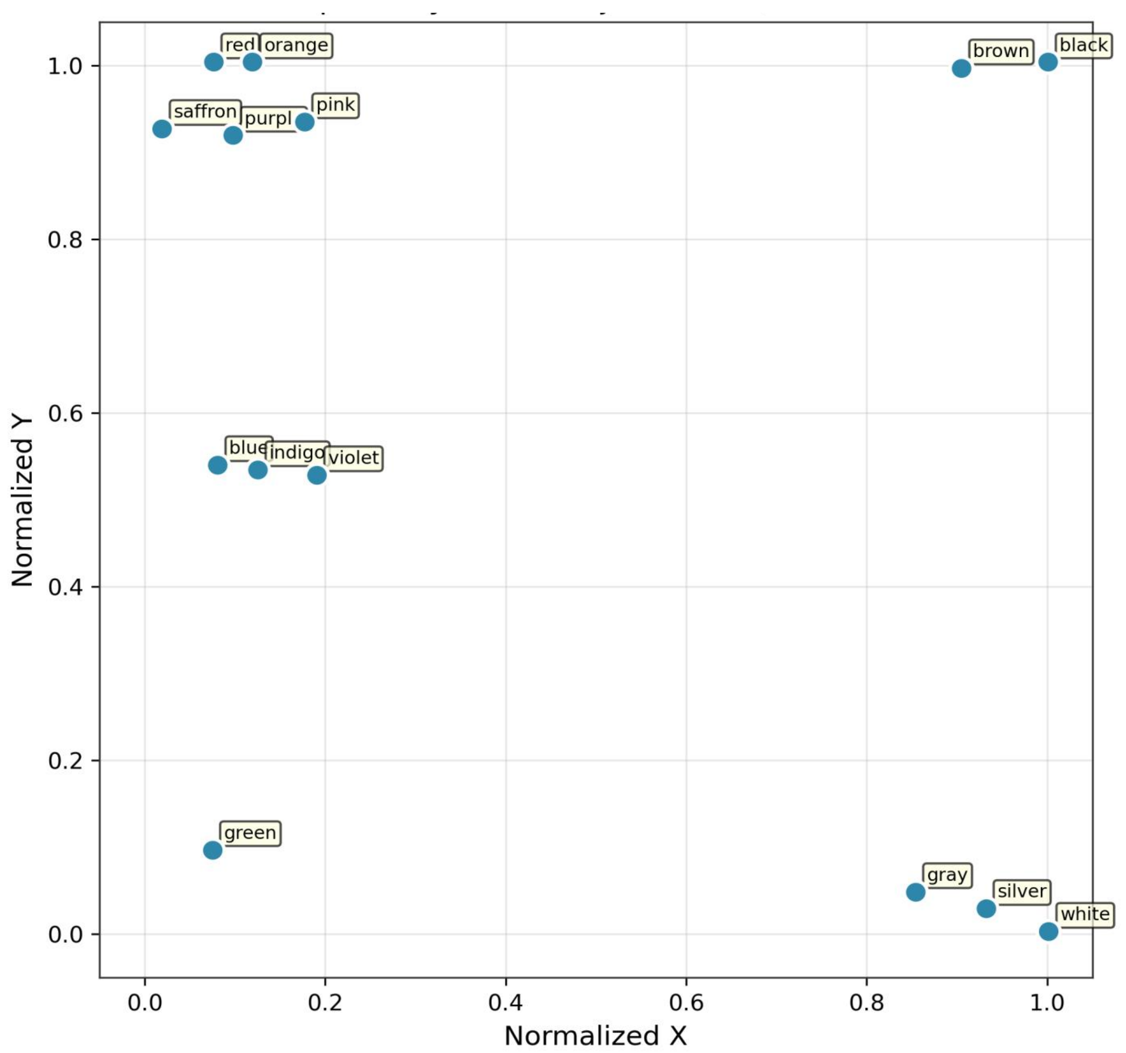
METHODOLOGY

Participants: 35 adults (20 native-Hindi, 15 other L1; ages 19–27)
Domains: animals, foods, body-parts, colours (each subject did 3 of 4)
Distance metrics (3 ways to measure "how far" two words are):

- Spatial – SpAM Euclidean (z-scored per trial)
- Phonological – Levenshtein on transliteration
- Semantic – multilingual embeddings (MiniLM,LaBSE, MuRIL)

Switch detection: SpAM-median & mean-split + Marginal-Value-Theorem (Hills et al.,2012)
Statistics: mixed-effects LMMs on log-IRT, subject random intercepts; Kruskal–Wallis across domains; paired Wilcoxon for the foraging spike

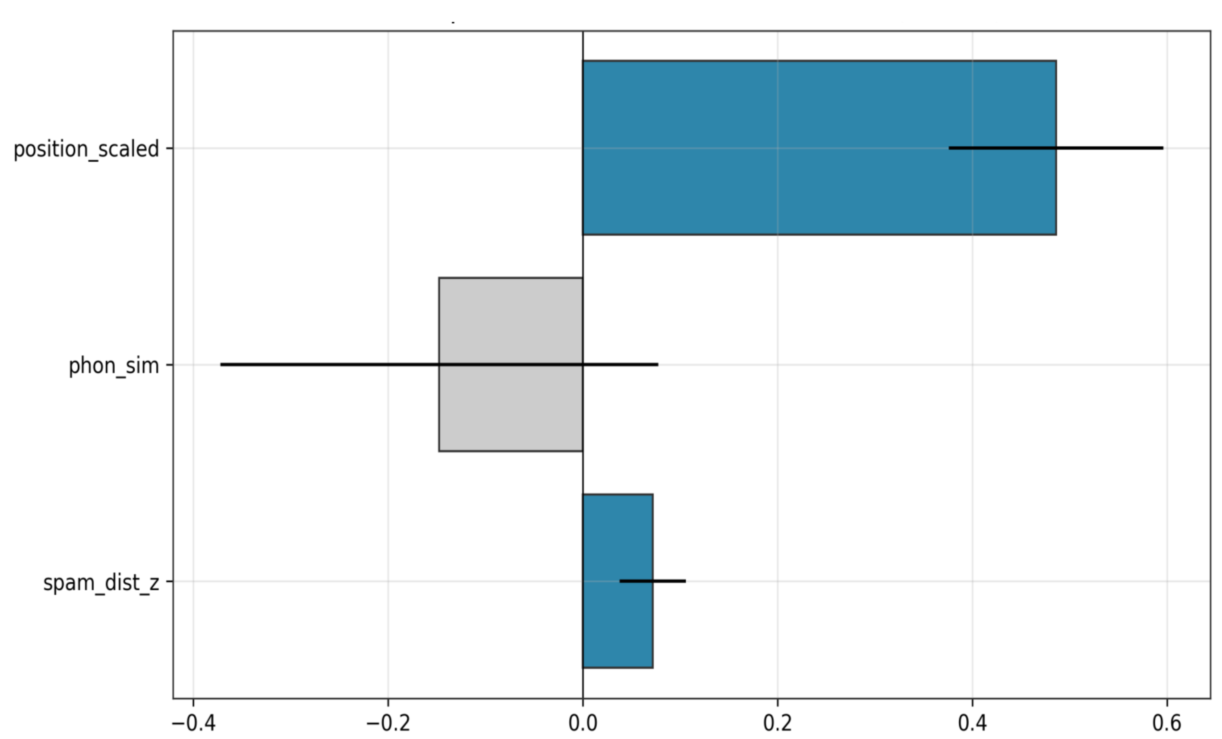
Fig. 1: Sample SpAM Arrangement (Subject 10255, Colours domain)



RESULTS

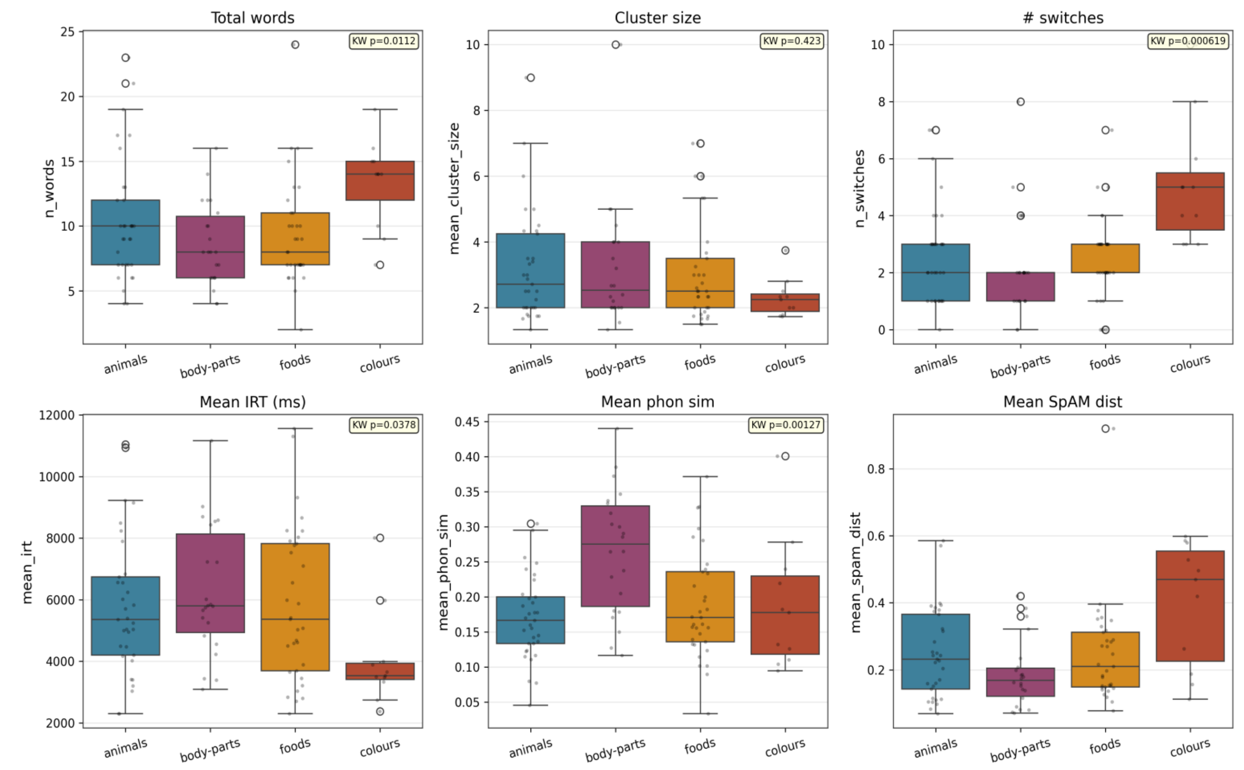
Joint cue model: SpAM + Phonology → IRT: SpAM distance significantly predicts log-IRT (mixed-effects: β = 0.097, p = 5.6 × 10−8), **resolving the Phase-1 Pearson null** (r = .026, p = .41). Robust across raw distance, winsorised IRT, and no-first-transition. Kumar et al. phonological prediction **not replicated**: phon similarity neither rises with position (β = −.009, p = .58) nor takes over at cluster exhaustion (phon × switch β = −.16, p = .51).

Fig. 2: Mixed-Effects Predictors of log-IRT (SpAM distance, phonology, position)



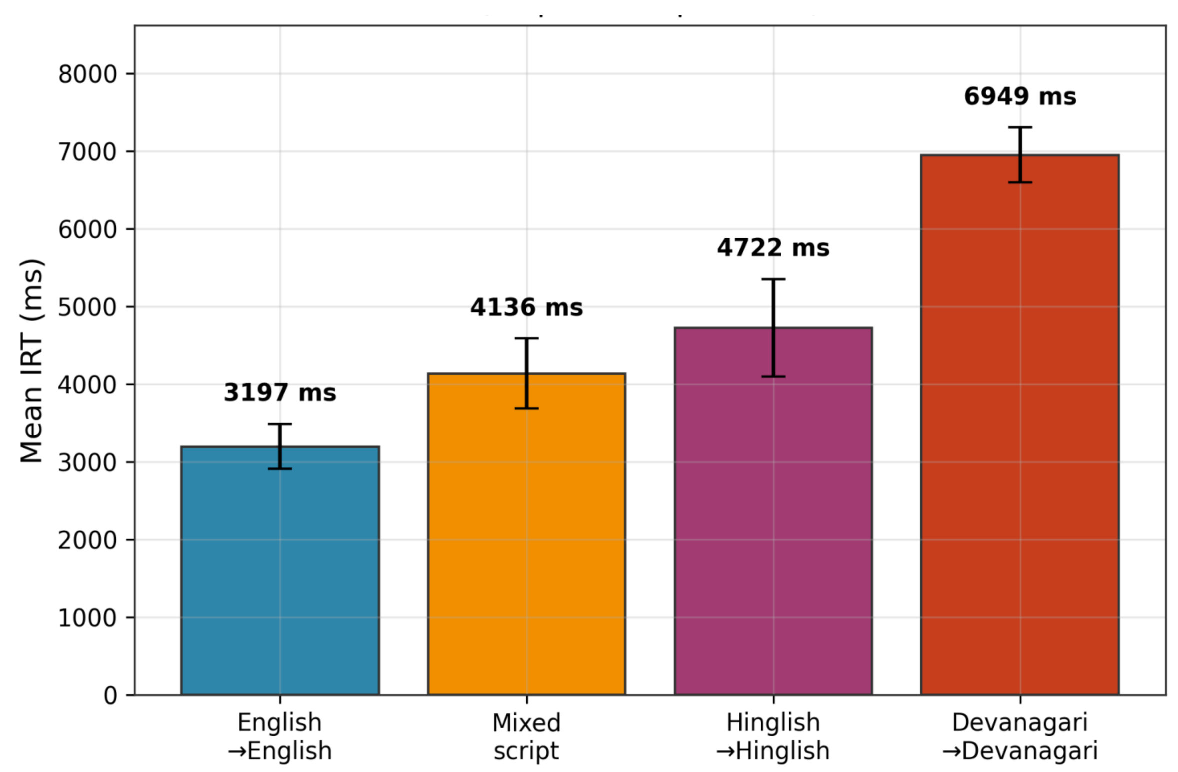
Domain differences & Fluency × Domain GLM: Kruskal–Wallis: domains differ on n_words (p = .011), IRT (p = .038), phon similarity (p = .001), and switches (p < .001); cluster size invariant (p = .42) – strategy is stable, pace and content vary. Fluency × body-parts interaction β = 0.219, p = .003.

Fig. 3: Retrieval Profile by Domain Word Count, Cluster Size, Switches, IRT, Phonology, SpAM Distance



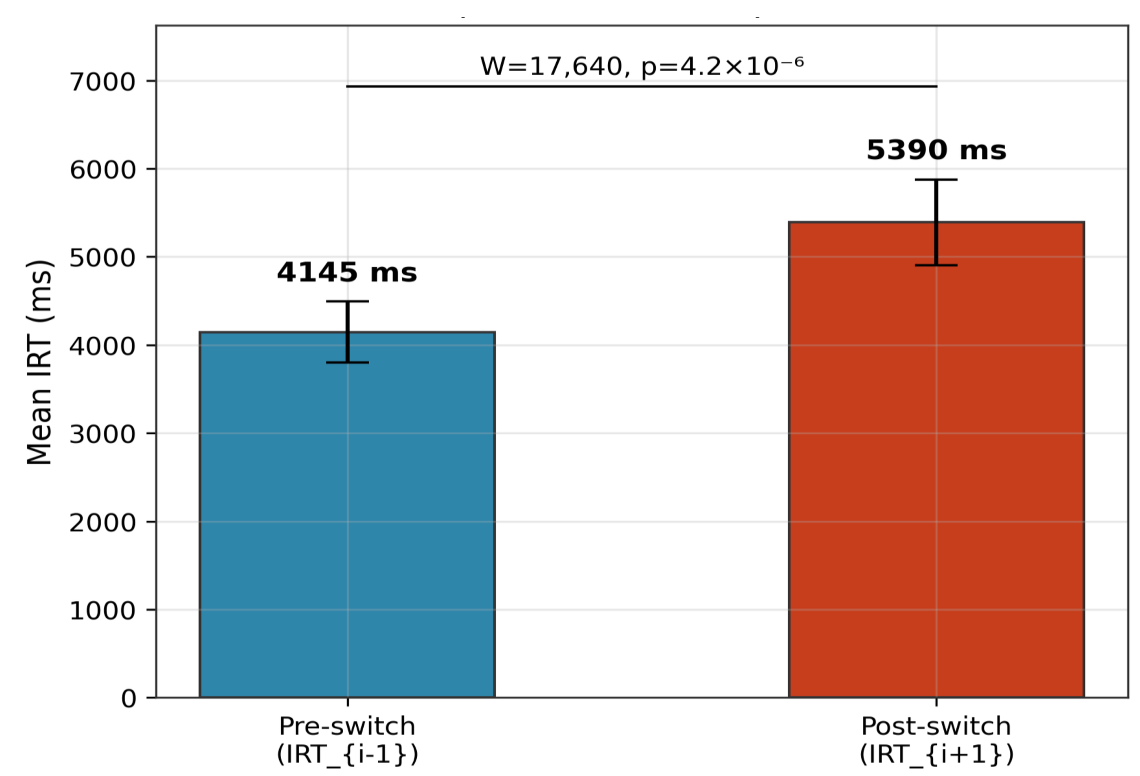
Script choice & the Devanagari typing penalty: Devanagari → Devanagari transitions are **2.2× slower** than English → English (6,949 vs. 3,197 ms; p < 10−53). Script preference is **not predicted** by L1 or any of Hi/En Read & Write Likerts (all p > .19) – script choice is idiosyncratic. The body-parts × fluency slowdown observed elsewhere is largely a typing-script confound, not retrieval depth.

Fig. 4: Devanagari Typing Penalty Mean IRT by Script Transition Type



Patch foraging – Marginal Value Theorem: Post-switch IRT **spike confirmed**: 5,390 vs. 4,145 ms (paired Wilcoxon W = 17,640, p = 4.2 × 10−6) – direct support for Hills, Jones & Todd (2012). MVT and SpAM-median switch detectors agree only 60% (χ2 = 4.9, p = .027): they capture different aspects of the foraging process.

Fig. 5: Post-Switch IRT Spike Direct Support for MVT (Hills et al., 2012)



CONCLUSION

- **Structured retrieval:** Hindi speakers search semantic memory in clusters; SpAM distance predicts IRT (β = 0.097, $p < 10^{-7}$), with position the dominant cue (β = 0.486).
- **Phonology is language-specific:** Unlike English, phonological similarity does not rise with position or rescue retrieval at cluster exhaustion — not a script artifact.
- **Domain shapes pace, not structure:** Word count, IRT, and switches vary across domains (all p < .05); cluster size is invariant (p = .42).
- **MVT confirmed:** Post-switch IRT spike directly supports optimal-foraging theory (5,390 vs 4,145 ms, $p < 10^{-5}$).
- **Typing penalty is a hidden confound:** Devanagari transitions are 2.2× slower than English ($p < 10^{-53}$); the body-parts fluency effect is largely a motor-output cost, not lexical depth.

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REFERENCES

•Kumar, Lundin & Jones (preprint). Mouse-mole-vole. PsyArXiv 2bazx_v1.
•Dautriche et al. (2016). Wordform ↔ semantic similarity, 100 languages. Cog Sci.
•Hills, Jones & Todd (2012). Optimal foraging in semantic memory. Psych Rev 119, 431–440.
•Troyer, Moscovitch & Winocur (1997). Clustering & switching in verbal fluency. Neuropsychology 11, 138–146.
•Feng et al. (2022) LaBSE; Khanuja et al. (2021) MuRIL.

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